

MTH204 (Section A)
Graded Worksheet–8

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

Q–1) Solve the IVP

$$y'' + 5y' + 6y = \delta\left(t - \frac{\pi}{2}\right) + u(t - \pi) \cos t \quad y(0) = 0, \quad y'(0) = 0$$

Q–2) Find the inverse Laplace transform of $\log \frac{s+a}{s+b}$.

Q–3) Find the Laplace transform of $3t \sinh 4t$.

Problem 4 and 5 are compulsory and solve any two from the rest three.

Q–4) Find the Laplace transform of the following functions:

1. $\cos^2(\omega t)$

2. $t^2 e^{-3t}$

3. $t \cos(4t)$

Q–5) Find the inverse Laplace transform of the following functions:

1. $\frac{5s+1}{s^2-25}$

2. $\frac{21}{(s+\sqrt{2})^4}$

3. $\frac{20}{s^3-2\pi s^2}$

Q–6) Solve the Initial Value Problem

$$y'' - y' - 6y = 0, \quad y(0) = 11, \quad y'(0) = 28.$$

Q–7) Solve the Initial Value Problem

$$y'' + 3y' - 4y = 6e^{2t-3}, \quad y(1.5) = 4, \quad y'(1.5) = 5.$$

Q-8) Solve the Initial Value Problem

$$y'' + 6y' + 8y = e^{-3t} - e^{-5t}, \quad y(0) = 0, \quad y'(0) = 0.$$

Solution:

Solution:

Solution:

Solution:

Solution: